

**Measures to Explore Biodegradable Plastic Alternatives as a Solution to Microplastic
Pollution in Southeast Asia's Coastal Ecosystems**

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Introduction

In modern society, microplastic pollution poses a severe threat to ecosystems around the globe. Microplastic refers to tiny plastic fragments (less than 5mm in size) produced by the breakdown of larger plastics. The negative impact of this pollution is prominent in the coastal ecosystems of Southeast Asia. Southeast Asia is an area filled with diverse and vigorous coastal ecosystems, including coral reefs and mangrove forests in the Philippines, Indonesia, Thailand, and Malaysia. However, due to the current microplastic pollution caused by single-use plastics, industrial waste, and the lack of waste collection, treatment, and management system, its coastal ecosystem is under an urgent threat. Moreover, the current microplastic pollution doesn't only damage the biodiversity of coastal ecosystems but also poses a risk to human health as humans consume the organisms living in the ecosystems. To resolve the current issue,



Map of Southeast Asia

various attempts have been made, and one of them is to utilize biodegradable plastic as an alternative source to traditional plastic. Biodegradable plastics can naturally decompose in a shorter amount of time while releasing less toxic and harmful materials. However, there are rising questions regarding the effectiveness of biodegradable plastics, demanding a practical and productive discussion to improve on and discover more solutions to the issue of marine microplastics.

Background

In the mid-20th century, after the Industrial Revolution, mass plastic production began in various industries, including packaging and healthcare. Due to the mass production of plastic, plastic slowly started to accumulate in the natural environment, especially in the form of microplastic. In the late 20th century, microplastic pollution was identified as a pivotal global issue as it damaged the world in various areas, including the environment, human health, and economics. First, microplastic pollution damages the ecosystems by harming marine organisms and causing bioaccumulation. As the toxic material carried by marine organisms was consumed by top predators, including humans, it posed a severe threat to



Microplastics found in Coastal Ecosystems

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human health, demanding an immediate approach to resolve the issue. Lastly, as microplastic pollution reduced the population of marine organisms and polluted beaches, it triggered tourist loss and damaged the fishing industries, negatively impacting the economy.

In Southeast Asian countries, microplastic pollution started in the late 1990s when plastic waste was transferred from developed to developing countries. With China as the main importer of the plastic waste trade,

Southeast Asian countries also started to receive plastic waste from developed countries, increasing microplastic pollution. However, in 2017, China issued a new policy banning the import of plastic waste, and from then, developed countries heavily relied on Southeast Asian countries for plastic waste trade, ultimately damaging the coastal ecosystems in Southeast Asia. Additionally, with the rapid economic



Plastic Waste in Southeast Asia

growth and urbanization in the late 20th century and early 21st century, plastic consumption substantially increased, and Southeast Asian countries lacked an adequate waste management system, leading to plastic leakage into coastal ecosystems.

Moreover, plastics play a significant role in Southeast Asian countries' economies, encumbering the actions to reduce plastic consumption and production. Plastics are widely used in Southeast Asia, mainly in areas of manufacturing and exporting, noting that Southeast Asia is the manufacturing hub for numerous companies. Additionally, transferring into biodegradable plastic alternatives presents a great economic challenge as biodegradable plastics are at a higher cost and require investments in new technologies. Lastly, Southeast Asian countries have heavily relied on single-use plastics and containers for the past decades, adding difficulty in reducing plastic consumption in Southeast Asia.

Problems Raised

Loss of Biodiversity

Due to plastic contamination, Southeast Asia is currently undergoing environmental degradation, especially in the coastal ecosystems. Microplastics that leak into the ocean damage the ecosystem by disrupting the food chain, damaging coral reefs, and leading to an overall loss in biodiversity. When organisms consume microplastic, they are unable to digest the manmade substance, and the microplastic then accumulates inside the digestive system, affecting the health of many animals and even leading to death in severe cases. As more and more organisms are exposed to the increasing levels of microplastic in the ocean, it becomes more likely that certain organisms face mass deaths. When there is such a change, other organisms higher on the trophic level are also affected due to their loss of food source, and this would lead to a turbulent food chain, possibly resulting in the disappearance of many organisms, diminishing the biodiversity of the coastal ecosystem. With coral reefs being home to one of the most diverse groups of organisms, the loss of diversity in these areas would have serious impacts on the health of the ecosystem.

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Water Pollution

The majority of the Southeast Asian countries leak massive amounts of plastic waste into the ocean, and if the trend continues, then the coastal ecosystems might turn into a

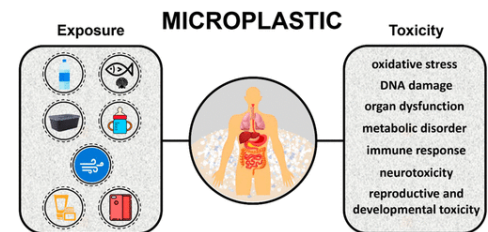
Table: The highest ocean plastic polluters

Rank	Country	Annual Ocean Plastic Waste (Metric tons)
#1	 Philippines	356,371
#2	 India	126,513
#3	 Malaysia	73,098
#4	 China	70,707
#5	 Indonesia	56,333
#6	 Myanmar	40,000
#7	 Brazil	37,799
#8	 Vietnam	28,221
#9	 Bangladesh	24,640
#10	 Thailand	22,806
	 Rest of the World	176,012
	Total	1,012,500

wasteland, limiting resources for the nations. For instance, the Philippines leaked 356, 371 metric tons of plastic waste into the ocean in 2021, accounting for 35% of global plastic waste leakage. Furthermore, Malaysia, Indonesia, Myanmar, Vietnam, Bangladesh, and Thailand have also leaked massive amounts of plastic waste into the ocean, ranking 3rd, 5th, 6th, 8th, 9th, and 10th in the world's highest ocean plastic polluters, respectively.

Health Risks

As humans consume the organisms living in the coastal ecosystems, humans may encounter health issues through biomagnification, such as ingestion of toxic chemicals. Although the exact effect of microplastic has not yet been declared, some potential risks include oxidative stress, DNA damage, cancers, and cognitive impairment.



Potential Health Risks posed by Microplastics

Negative Economic Effects

Due to microplastic pollution, numerous organisms have died in the coastal ecosystem, and this negatively affected the coastal communities heavily relying on fishing and tourism. For instance, due to microplastic pollution, Boracay, an island in the Philippines, lost tourists and had to temporarily close the island to clean up the region. Moreover, Maya Bay, Thailand, and Bali, Indonesia, had to take similar approaches to further sustain their economy by cleaning up the region to attract tourists.

Concerns Regarding the Feasibility of Biodegradable Plastics

Biodegradable plastics can only decompose efficiently under specific environmental conditions depending on their temperature and humidity. However, in Southeast Asia, industrial facilities to control the environmental conditions and manage the plastics are limited, thus lacking the feasibility of replacing traditional plastics with biodegradable plastic alternatives. Moreover, biodegradable plastics require a well-managed system, and if mismanagement occurs and the biodegradable plastics are treated like traditional plastics, then their decomposition rate may be significantly slower or halted. Additionally, consumers may misunderstand the concept of biodegradable plastics and believe that they can

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decompose under any environmental situation, which may lead to increased littering and inappropriate disposal.

Higher Costs of Biodegradable Plastics



Corns used to produce Biodegradable Plastics

In comparison with traditional plastics, biodegradable plastics are at a higher cost, restricting them from being widely adopted around the nation. For instance, a biodegradable plastic called PLA (Polylactic Acid) costs around 3\$ per kilogram, whereas a non-biodegradable plastic (traditional plastic) called PET (Polyethylene Terephthalate) costs around 1\$ per kg. Furthermore, the majority of Southeast Asian countries are Less Economically Developed Countries (LEDCs), meaning that an effective and

supportive approach needs to be taken to resolve the issue. Additionally, the majority of biodegradable plastics require agricultural resources, mainly corn and sugarcane, to be produced, and this may lead to deforestation and higher greenhouse gas emissions due to overcultivation.

International Actions

United Nations Initiatives

The United Nations Environment Programme (UNEP) launched a campaign called “Clean Seas”, focusing on reducing microplastic pollution in oceans. Also, in the Sustainable Development Goals (2030 Agenda) proposed by the UNEP, goal 14, Life Under Water, specifically aims to recover the coastal ecosystems and reduce marine pollution. However, recent research produced by the UNEP stated that biodegradable plastics are not the best solution to marine litter and ecosystem degradation.



The UN 2030 Agenda for Sustainable Development

Private Corporations

Private companies, mainly Nestlé and Unilever, have aimed to create biodegradable packaging for their products, thus further investing in finding biodegradable plastic alternatives. Also, The Bioplastic Feedstock Alliance (BFA) mainly invests in creating biodegradable plastics to sustain our environment.



Biodegradable bottle packaging produced by Unilever

Regional Agreements

In 2019, the Association of Southeast Asian Nations (ASEAN) agreed to follow the Bangkok Declaration on Marine Debris, aiming to recover the coastal ecosystems and reduce

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microplastic pollution. In addition to the Bangkok Declaration on Marine Debris, the ASEAN also released the Blue Economy Framework, which emphasized cross-country cooperation in reducing marine debris. Furthermore, ASEAN countries decided to deeply explore alternative materials, including biodegradable plastics, to resolve the current issue of microplastic pollution in coastal ecosystems.

Major Parties Involved

The Association of Southeast Asian Nations

ASEAN is an association targeted toward promoting peace and stability in Southeast Asian nations, as well as the economic and cultural exchange among member states. As microplastic pollution is currently occurring in Southeast Asian coastal ecosystems, ASEAN plays a pivotal role in resolving the issue. As the Association of Southeast Asian Nations, ASEAN promotes cooperation between nations to reduce microplastic pollution and explore biodegradable plastic alternatives. Furthermore, ASEAN has already agreed to the Bangkok Declaration on Marine Debris, aiming to reduce marine debris and microplastic pollution in oceans. Without ASEAN, the cooperation between Southeast Asian countries will face difficulty, thus ASEAN plays a crucial role in resolving the current issue.

United Nations Environment Programme

As the microplastic pollution is currently severely damaging the environment, UNEP supports Southeast Asian countries and provides funding to explore biodegradable plastic as an alternative source to traditional plastics. Moreover, the UNEP provides policy guidance for researching biodegradable plastics, ensuring that nations can effectively discover alternative sources in a timely manner. Without the UNEP, international cooperation will face difficulty, thus it plays a pivotal role in resolving the current issue.

Bioplastic Feedstock Alliance

Bioplastic Feedstock Alliance, or BFA, is a multi-stakeholder initiative that leads the production of biodegradable plastics from renewable resources. They play a significant role in this issue as their development of biodegradable plastics may act as an alternative source to replace traditional plastics, which will ultimately reduce microplastic pollution.



National University of Singapore

The National University of Singapore, or NUS, is the largest research institution in Southeast Asia. As such, it aims to study the impact of biodegradable plastics and explore alternative materials to replace traditional plastics. In addition, NUS provides scientific assistance in reducing microplastic pollution in coastal ecosystems, thus acting as a key player in resolving the current issue.

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Possible Solutions

Development of Monitoring Technology

The fundamental way of solving this issue would be cleaning up the ocean using various waste management and cleaning methods. To maintain the reduced concentration of microplastic in the bodies of water, consistent monitoring of the concentration levels is also crucial, and this can be completed by running samples of water under Raman spectroscopy or FTIR spectroscopy. A simple way would be collecting multiple samples in certain time intervals and performing spectroscopy with the samples to record concentration levels over time. For this to be possible, nations surrounded by high-risk areas of microplastic pollution need to be equipped with sufficient technology and professionals, and to make this testing system more efficient and manageable, efforts need to be put in to developing effective monitoring technology.

Restoration of Biodiversity

As one of the major ecological harms due to microplastic pollution, the loss of biodiversity requires efforts to be put forth to restore the diversity of species in the affected marine regions. A major method to this is to introduce the lost and damaged species back into the ecosystems after sufficient cleaning efforts. For instance, in affected coral reefs, artificial coral reefs and replantation of corals can be used to restore the coral reef habitat, which is suitable for a wide spectrum of species. If a key organism's population was lost due to microplastic pollution, reintroducing groups of this species into the ecosystem will help them proliferate and regrow their population size. Ultimately, this would then help bring back the numbers of other related species, helping the ecosystem regain its stability.

Investing in Biodegradable Plastics

Investing in biodegradable plastic will allow research institutions to accelerate their research pace and effectively develop a type of biodegradable plastic suitable to the Southeast



A Scientist researching on Biodegradable Plastics

Asian climate, which ultimately will resolve the current microplastic pollution. Another method of investing in biodegradable plastics is to provide incentives to corporations utilizing biodegradable plastics; however, to bring this into reality, cooperation with financial organizations will be necessary. Furthermore, investing in the creation of industrial facilities to decompose biodegradable plastics will also increase the feasibility and effectiveness of biodegradable plastics.

Strengthening Policies

Strengthening policies regarding this issue, such as banning the usage of single-use plastics and establishing a certification system for biodegradable plastics, will further

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enhance the current situation. By banning the usage of single-use plastics, plastic waste leakage will substantially decrease as the majority of plastic waste derives from single-use plastics consumed by individuals. Establishing a certification system for biodegradable plastics will also enhance the situation by reducing confusion and ensuring the quality of biodegradable plastics.

Raising Public Awareness

Numerous individuals living in Southeast Asian countries are not fully aware of the current issue, thus an immediate approach to educate citizens and alert the urgency of the situation is demanded. If the citizens are unaware of the issue, then the usage of non-biodegradable plastics will continue, encumbering it to reduce microplastic pollution. Examples of raising public awareness may include educating consumers on the benefits and disadvantages of biodegradable plastics, informing citizens about the urgency of coastal ecosystem degradation, establishing campaigns to reduce plastic usage, and forming funds to support biodegradable plastics research.



A Man Raising Public Awareness on Microplastic Pollution

Glossary

Bioaccumulation: The accumulation of substances, especially toxic chemicals, in an organism.

Biodegradable Plastics: Plastics that can decompose efficiently under certain environmental conditions depending on their temperature and humidity.

Biodiversity: The variety of species living in a certain ecosystem.

Coastal Ecosystems: A marine ecosystem where the land meets the sea; examples include coral reefs and mangrove forests.

Deforestation: The action of cutting down trees for agricultural or commercial purposes in a large area.

Marine Debris: Manufactured objects, such as plastics, metals, and rubbers, released in seas or oceans (marine ecosystems).

Microplastic: Tiny plastic fragments (less than 5mm) produced by the breakdown of larger plastics.

Overcultivation: The action of excessively cultivating certain crops.

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Timeline

1950 – Proliferation of plastic production begins.

1970 – Microplastic is identified as a severe environmental issue for the first time.

1990 – Plastic waste starts to transfer from developed countries to developing countries.

1995 – The Southeast Asian countries experience rapid economic growth and urbanization.

2010 – Southeast Asian countries are identified as the major source of marine plastic pollution.

2019 – The Association of Southeast Asian Nations (ASEAN) agrees to the Bangkok Declaration on Combating Marine Debris

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